

## The Relativistic Particle:

$$S = -m \int_{t_0}^{t_f} ds = -m \int_{t_0}^{t_f} dt \sqrt{-\frac{dx^\mu}{dt} \frac{dx^\nu}{dt} \eta_{\mu\nu}}$$

i.e.

$$\gamma: \tau \mapsto X^\mu(\tau) \in \mathbb{M}^{1,d-1}$$

(Nambu-Goto action)

$$S \equiv \int d\tau L(\tau)$$

$\mu=0, i$

• If we identify  $\tau$  w/ the

proper time:

$$-\frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \eta_{\mu\nu} = 1 - \vec{v}^2$$

$$S = -m \int d\tau \sqrt{1 - \vec{v}^2}$$

Recall that  $X^\mu = (t, \vec{x})$

$$\frac{dt}{d\tau} = \gamma = (1 - \vec{v}^2)^{-1/2}$$

$$\text{and so } \frac{dx^\mu}{d\tau} = (\gamma, \gamma \vec{v}) = \gamma(1, \vec{v})$$

$$\therefore \left(\frac{dx^\mu}{d\tau}\right) \frac{dx^\nu}{d\tau} \eta_{\mu\nu} = -\gamma^2 + \vec{v}^2 \gamma^2 = -1$$

\* Recall that the proper time is

$$-d\tau^2 = ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

Often, texts will use  $\int ds$  when really this is  $d\tau$ .

$$\Leftrightarrow d\tau/dt = \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

Since  $X^\mu = (t, \vec{x})$

$$d\tau/dt = (1 - \vec{v}^2)^{1/2}$$

$$\Leftrightarrow \boxed{\frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \vec{v}^2}} = \gamma}$$

i.e.

hence

\* Diffeomorphisms: let  $\tau \rightarrow \tilde{\tau}(\tau)$

then

$$\frac{dx^\mu}{d\tilde{\tau}} = \frac{dx^\mu}{d\tau} \frac{d\tau}{d\tilde{\tau}}$$

$$d\tilde{\tau} = \frac{\partial \tilde{\tau}}{\partial \tau} d\tau$$

$$\tilde{S} = -m \int d\tilde{\tau} \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tilde{\tau}} \frac{dx^\nu}{d\tilde{\tau}}}$$

$$= -m \int \left( \frac{\partial \tilde{\tau}}{\partial \tau} d\tau \right) \sqrt{-\eta_{\mu\nu} \left( \frac{dx^\mu}{d\tau} \frac{\partial \tau}{\partial \tilde{\tau}} \right) \left( \frac{dx^\nu}{d\tau} \frac{\partial \tau}{\partial \tilde{\tau}} \right)}$$

$$= -m \int d\tau \left( \frac{d\tilde{\tau}}{d\tau} \right) \left( \frac{d\tau}{d\tilde{\tau}} \right) \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

$$= S \quad \checkmark$$

\* From this we are free to parametrize  $\tau$  as we wish & we often choose affine coords.

Second order formalism: Introduce  $e_{\text{rel.}}$  or the Polyakov action for massive point particles in  $ND$   $e^2 \equiv g_{\tau\tau}$

$$S_p \equiv \frac{1}{2} \int d\tau e(\tau) \left( \frac{1}{e^2(\tau)} \left( \frac{dx^\mu}{d\tau} \right)^2 - m^2 \right)$$

$$\bullet \dot{x}^2 \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$$

\* Proof that  $S_p = S_{NG}$ :

$$\frac{\delta S_p}{\delta e} = 0 \Leftrightarrow -\frac{1}{e^2} (\dot{x})^2 - m^2 = 0$$

$$e^2 = -\frac{\dot{x}^2}{m^2}$$

$$\text{and } \frac{\delta S_p}{\delta x^\mu(\tau)} = 0 \Leftrightarrow g_{\mu\nu} \dot{x}^\nu = \frac{d}{d\tau} \left( \frac{1}{e} \delta_{\mu\nu} \dot{x}^\nu \right) \left( \frac{dx^\mu}{d\tau} \right)$$

$$= \int d\tau 2 \delta(\tau - \tau') \frac{d}{d\tau} (e^{-1}(\tau) \dot{x}^\mu) = 0$$

$$\frac{d}{d\tau} (e^{-1} \dot{x}^\mu) = 0$$

Since E.O.M. of  $e$  are algebraic it has no dynamics and is purely aux.

The Canonical Momenta:  $P^\mu := \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

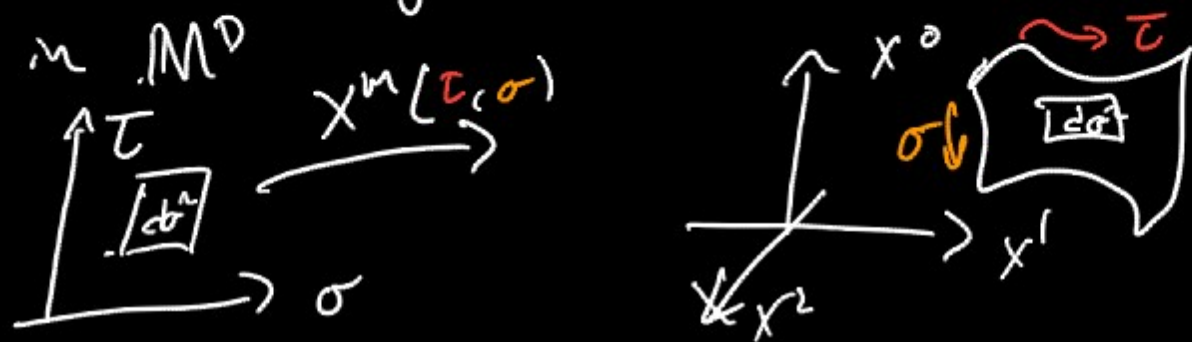
Here  $P^\mu = \dot{x}^\mu(\tau)$

For  $e^2 = -\dot{x}^2/m^2 |e=1$  we simply have the mass-shell condition

$$\boxed{P_\mu P^\mu + m^2 = 0}$$

The Bosonic String:

• We can, topologically, only have open or closed strings. These will be embedded in  $M^D$



$\Sigma$ : World sheet

Which is parametrized by  $\tau \in \mathbb{R}, 0 \leq \sigma \leq \ell$

$\Sigma: (\tau, \sigma) \mapsto x^\mu(\tau, \sigma) \in M^D$

which we define to be the string map.

$\Sigma \hookrightarrow M^D$

We use this to find the action of the string. Above, we calculated

the minimum of the length of  $\sigma$  defined by a particle. Here we minimize the Area of the string.

Let  $\sigma^a \equiv (\tau, \sigma)$  be worldsheet coords  
 $\sigma^0 = \tau, \sigma^1 = \sigma$

then we have

$$X : \Sigma \rightarrow M^D$$

$$(\tilde{X})^* : T^*M^D \rightarrow T^*\Sigma$$

$$\begin{aligned} (\tilde{X})^* \eta((X, Y)) &= \eta(\tilde{X}^* X, \tilde{X}^* Y) \\ &= (X^\alpha \partial_\alpha \tilde{X}^\mu \partial_\mu, Y^\beta \partial_\beta \tilde{X}^\nu \partial_\nu) \\ \Leftrightarrow (\tilde{X}^* \eta)_{\alpha\beta} &= \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \end{aligned}$$

$\therefore$  The metric on  $\Sigma$  is given by

$$G_{\alpha\beta} : T\Sigma \times T\Sigma \rightarrow \mathbb{R}$$

$$G_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X^\nu$$

$$S_{NG} = -T \int dA = -T \int \sqrt{\det h_{\alpha\beta}} d^2\sigma$$

$$G_{00} = \dot{X}^2$$

$$G_{11} = \frac{dX^\mu}{d\sigma} \frac{dX^\nu}{d\sigma} \equiv X'^2$$

$$G_{01} = \frac{dX^\mu}{d\tau} \frac{dX^\nu}{d\sigma} \eta_{\mu\nu}$$

$$[G_{\alpha\beta}] = \begin{pmatrix} \dot{X}^2 & \frac{dX^\mu}{d\sigma} \frac{dX^\nu}{d\tau} \eta_{\mu\nu} \\ \frac{dX^\mu}{d\sigma} \frac{dX^\nu}{d\sigma} \eta_{\mu\nu} & X'^2 \end{pmatrix}$$

$$\det(\cdot) = \dot{X}^2 X'^2 - (\dot{X} \cdot X')^2$$

$$S_{NG} = -\frac{T}{2} \int d\tau d\sigma \sqrt{(\dot{X} \cdot X')^2 - (\dot{X}^2) (X')^2}$$

worldsheet metric  $h_{\alpha\beta}(\tau, \sigma)$  which is different from the pull back  $G_{\alpha\beta}$

Polyakov Action:

$$S_P := -\frac{1}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta}(\sigma) \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

$$= -\frac{1}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} G_{\alpha\beta}$$

\* For a spacetime  $M^D$  w/ metric  $g_{\mu\nu}$  we simply replace  $\eta_{\mu\nu}$  w/  $g_{\mu\nu}(X)$ . This leads to a non-linear sigma-model.

$$\rightarrow g_{\mu\nu}(X) = \eta_{\mu\nu}$$

Can be interpreted as a lowest order expansion  $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} + \mathcal{O}(h^2)$

$$\delta S_P = -\frac{1}{2} \int d^2\sigma \delta(\sqrt{-h} h^{\alpha\beta}) \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}$$

$$= -\frac{1}{2} \int d^2\sigma \left[ \left( -\frac{1}{2} (\sqrt{-h} h^{\gamma\delta} \delta h^{\gamma\delta}) h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \right. \right.$$

$$\left. \left. + \sqrt{-h} \delta h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} \right] \right]$$

$$= -\frac{1}{2} \int d^2\sigma (\delta h^{\gamma\delta}) \sqrt{-h} \left[ \frac{1}{2} h_{\gamma\delta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} + \partial_\gamma X^\mu \partial_\delta X^\nu g_{\mu\nu} \right]$$

The energy-momentum tensor:

$$T_{\alpha\beta} := \left( -\frac{2}{\sqrt{-h}} \right) \frac{\delta S}{\delta h^{\alpha\beta}}$$

$$= -\frac{1}{2} \int d^2\sigma \sqrt{-h} (\delta h^{\gamma\delta}) \left( \frac{1}{2} h_{\gamma\delta} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu} + \partial_\gamma X^\mu \partial_\delta X^\nu g_{\mu\nu} \right)$$

$$T_{\alpha\beta} = -\frac{1}{2} h_{\alpha\beta} h^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X^\nu g_{\mu\nu} + \partial_\alpha X^\mu \partial_\beta X^\nu g_{\mu\nu}$$

that the  $\gamma$  must be reabsorbed in order to have  $(h)/\sqrt{-h} = \sqrt{-h}$ ; i.e.

$$\delta \sqrt{-h} = \left(-\frac{1}{2}\right) \frac{1}{\sqrt{-h}} \delta h = \left(\frac{1}{2} \frac{1}{\sqrt{-h}}\right) (-h) (\delta h^{\alpha\beta})$$

$$= -\frac{1}{2} \sqrt{-h} h_{\alpha\beta} \delta h^{\alpha\beta}$$

E.O.M for  $h^{\alpha\beta}$ , i.e. (second order formalism):

$$T_{\alpha\beta} = 0 \Leftrightarrow \frac{1}{2} h_{\alpha\beta} h^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X_\mu = \partial_\alpha X^\mu \partial_\beta X_\mu$$

• On-shell, the pull-back metric  $\nearrow$  and the induced metric  $h_{\alpha\beta}$  are proportional.

$$G_{\alpha\beta} = h_{\alpha\beta} \left(\frac{1}{2} h^{\gamma\delta} G_{\gamma\delta}\right)$$

\* E.O.M for  $X^\mu$ :

$$\delta S = -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu$$

$$= +T \int d^2\sigma \partial_\alpha (\sqrt{-h} h^{\alpha\beta} \partial_\beta X^\mu) \delta X_\mu$$

$$\Rightarrow \Delta X^\mu = \frac{1}{\sqrt{-h}} \partial_\alpha (\sqrt{-h} h^{\alpha\beta} \partial_\beta X^\mu) = 0$$

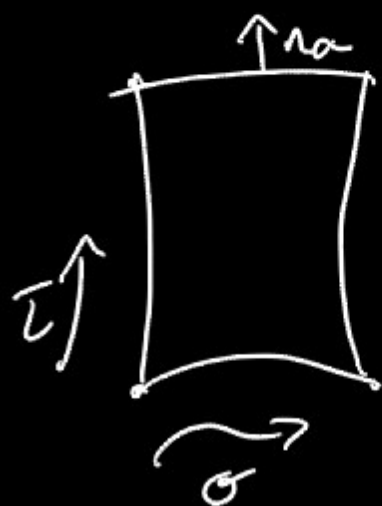
which has B.C.'s:

(i) Closed String:

$$X^\mu(\tau, \sigma + l) = X^\mu(\tau, \sigma)$$

(ii) Open String:

$$\eta^\alpha \partial_\alpha X^\mu \delta X_\mu |_{\sigma=0,l} = 0$$



$T_{\alpha\beta}$  is covariantly constant:

$$\nabla_\alpha T^{\alpha\beta} = 0$$

\* The Polyakov metric is not, like NG action, the Area of the worldsheet.

Area:  $\int d^2\sigma \sqrt{-h}$

Let  $A$  be a symm.  $\in M_{2 \times 2}(\mathbb{R})$ . Then

$$(\text{tr}(A))^2 \geq 4 \det(A)$$

Proof:  $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$ ,  $\det(A) = \prod_{i=1}^n \lambda_i$

then AM-GM inequality states

$$\frac{x_1 + \dots + x_n}{n} \geq (x_1 \dots x_n)^{1/n}$$

$$\Rightarrow A \in M_{2 \times 2}(\mathbb{R}) \Leftrightarrow n=2$$

$$\frac{(\lambda_1 + \lambda_2)}{2} \geq (\lambda_1 \lambda_2)^{1/2}, \text{ or}$$

$$(\lambda_1 + \lambda_2)^2 \geq 4 \lambda_1 \lambda_2$$

$$\text{Tr}(A)^2 \geq 4 \det(A)$$

Equality holds for  $A \propto \mathbb{I}$ .

\* For  $A^\alpha_\beta = h^\alpha_\gamma G_{\gamma\beta} \Leftrightarrow S_p \geq S_{NG}$ .

Equality holds when  $G_{\gamma\beta}$  is conformally eq.

to  $h_{\alpha\beta}$ .  $\Leftrightarrow$  On-shell  $T_{\alpha\beta} = 0$  condition implies

conformal equivalence of the pull back and

$h_{\alpha\beta}$ .

Q: Are there other terms which we could add to the string action?

(i) Conformal anomaly term can be added

$$S_1 = \Lambda_1 \int d^2\sigma \sqrt{-h}$$

• We must have  $\Lambda_1$  to have Weyl invariance.

(ii) 2D Einstein-Hilbert term

$$S_2 = \frac{\Lambda_2}{4\pi} \int d^2\sigma \sqrt{-h} R^{(2)}$$

This has no dynamics as 2D gravity is trivial

$$\chi(\Sigma) = \frac{1}{4\pi} \int d^2\sigma \sqrt{-h} R^{(2)} \quad (\text{Euler characteristic})$$

(For closed string)

$$R^{(2)} \sqrt{-h} \sim \partial_\alpha ( \quad )$$

The Polyakov action for propagation of a string in  $\mathbb{R}^{1,d-1}$  describes  $d$  two dimensional scalar fields  $x^\mu(\sigma)$  coupled to the dynamical WS metric  $h_{ab}(\sigma)$

→ Studying Bosonic ST is equivalent to studying 2-d gravity coupled to scalars

Polyakov symmetries:

We must distinguish between Spacetime on  $\mathbb{M}^d$  and the WS symm. acting on  $\Sigma$ .

i) Spacetime symmetries:

We have Poincaré invariance

$$x^\mu(\sigma) \mapsto \Lambda^\mu{}_\nu (x^\nu - a^\nu), \quad \Lambda \in SO(1, d-1)$$

Recall:  $(M_{[pq]})^\mu{}_\nu := \xi^\mu_p \eta_{\nu q} - \delta^\mu_q \eta_{\nu p}$

$$\delta h_{\alpha\beta} = 0$$

$$\text{Diff}(\Sigma_g) \equiv \{f: \Sigma_g \rightarrow \Sigma_g\}$$

$$f^* X^\mu$$

$$f^* h_{\alpha\beta} = \frac{\partial \xi^\sigma}{\partial \xi'^\alpha} \dots$$

(local) WS symmetries:

→ Reparametrization invariance

$$\sigma \mapsto \sigma' := \sigma + \xi^\alpha(\sigma)$$

$$(i) \text{ Scalar: } X^\mu(\sigma) \mapsto X'^\mu(\sigma') = X^\mu(\sigma(\sigma'))$$

$$X'^\mu(\sigma') = X^\mu(\sigma') + \xi^\alpha \partial_\alpha X^\mu / \sigma$$

$$\Rightarrow \delta X^\mu = X'^\mu(\sigma) - X^\mu(\sigma) = \xi^\alpha \partial_\alpha X^\mu(\sigma)$$

(ii) Metric:

$$\delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha$$

(iii) Fun:

$$\delta \sqrt{-h} = \frac{1}{2} h^{\alpha\beta} (\xi^\gamma \partial_\gamma h_{\alpha\beta} + \partial_\alpha \xi^\gamma h_{\gamma\beta} + \partial_\beta \xi^\gamma h_{\alpha\gamma})$$

$$\text{note } \partial_\alpha \sqrt{-h} = -\frac{1}{2} \frac{1}{\sqrt{-h}} h^{\gamma\delta} \partial_\alpha h_{\gamma\delta}$$

$$\Rightarrow \delta \sqrt{-h} = \partial_\alpha (\xi^\alpha \sqrt{-h})$$

$$* \text{ For } \Lambda^\mu_\nu = \delta^\mu_\nu + \lambda^\mu_\nu$$

$$\Rightarrow \delta X^\mu = \lambda^\mu_\nu X^\nu$$

To see that  $\Lambda$  is antisymmetric:

$$M_{\mu\nu} X^\mu X^\nu = M_{\mu\nu} X'^\mu X'^\nu$$

$$\Rightarrow \delta(M_{\mu\nu} X^\mu X^\nu) = 0$$

$$2 M_{\mu\nu} \lambda^\mu_\sigma X^\sigma X^\nu = 0$$

$$2 \underbrace{\lambda_{\nu\sigma}}_{\text{Antisym}} \underbrace{X^\sigma X^\nu}_{\text{Sym}} = 0$$

i) Rotation about  $x^3$  by an angle  $\theta$

$$x'^1 = \cos(\theta) x^1 + \sin(\theta) x^2$$

$$x'^2 = -\sin(\theta) x^1 + \cos(\theta) x^2$$

$$\theta \ll 1, \cos \rightarrow 1 \quad \sin \rightarrow \theta$$

$$x'^1 = x^1 + \theta x^2$$

$$x'^2 = -\theta x^1 + x^2$$

$$\delta x^1 = \theta x^2, \quad \delta x^2 = -\theta x^1$$

$$\text{but since } \delta x^m = a^m_n x^n$$

$$a^1_2 = \theta, \quad a^2_1 = -\theta \Rightarrow a_{12} = \eta_{11} a^1_2 = \theta$$

(ii) Boost in the  $x^0$  &  $x^1$  direction by  $\varphi$ .

$$x'^0 = \cosh(\varphi) x^0 + \sinh(\varphi) x^1$$

$$x'^1 = \sinh(\varphi) x^0 + \cosh(\varphi) x^1$$

$$\cosh \rightarrow 1, \quad \sinh \rightarrow \varphi$$

$$\delta x^0 = \varphi x^1, \quad \delta x^1 = \varphi x^0$$

$$\Rightarrow \lambda^0_1 = \varphi, \quad \lambda^1_0 = \varphi$$

Invariance of  $S_p$  under Poincaré:

$$\delta S = -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha (\delta x^\mu) \partial_\beta x^\nu g_{\mu\nu}$$

$$= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha (\lambda^\mu_\sigma x^\sigma + a^\mu) \partial_\beta x^\nu g_{\mu\nu}$$

$$= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \lambda^\mu_\sigma \partial_\alpha x^\sigma \partial_\beta x^\nu g_{\mu\nu}$$

$$= -T \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \underbrace{\lambda_{\sigma\nu}}_{\text{Anti}} \underbrace{\partial_\alpha x^\sigma \partial_\beta x^\nu}_{\text{Sym}}$$

$$\therefore \delta S = 0 \quad (\text{Poincaré invariant})$$

## \* World Sheet (local) Symmetries:

$$h_{\alpha\beta} \mapsto h'_{\alpha\beta} \equiv e^{2\phi(\sigma)} h_{\alpha\beta}(\sigma)$$

$$\Rightarrow \delta h_{\alpha\beta} = 2\phi(\sigma) h_{\alpha\beta}$$

$$\delta x^m = 0 \quad \forall m.$$

(:) We note that  $\sqrt{-h}$  is a tensor density

and transforms exactly as  $h_{\alpha\beta}$

$$\begin{aligned}\sqrt{-h'} &= \sqrt{-\det(e^{2\phi} h_{\alpha\beta})} = \sqrt{-e^{4\phi} h} \\ &= e^{2\phi} \sqrt{-h}\end{aligned}$$

\* The inverse of  $h$  transforms as

$$h^{\alpha\beta} \mapsto e^{-2\phi} h^{\alpha\beta}$$

The pair  $\sqrt{-h} h^{\alpha\beta}$  has conformal weight 0:

$$\sqrt{-h'} h'^{\alpha\beta} = e^{2\phi} \sqrt{-h} e^{-2\phi} h^{\alpha\beta} = \sqrt{-h} h^{\alpha\beta} \quad \checkmark$$

Claim: Conf. invariance implies the E.M tensor is traceless:  $h^{\alpha\beta} T_{\alpha\beta} = 0$

Proof:  $T_{\alpha\beta} = -\frac{2}{T} \frac{1}{\sqrt{-h}} \frac{\delta S}{\delta h^{\alpha\beta}}$ , thus

$$\delta S = \int \frac{\delta S}{\delta h^{\alpha\beta}} \delta h^{\alpha\beta} d^2\sigma$$

$$= -\frac{T}{2} \int d^2\sigma \sqrt{-h} T_{\alpha\beta} \delta h^{\alpha\beta}$$

$$= T \int d^2\sigma \sqrt{-h} T_{\alpha\beta} (\phi) h^{\alpha\beta} \stackrel{!}{=} 0$$

Since  $\phi \neq 0$  and  $\sqrt{-h} \geq 0$  we find

$$h^{\alpha\beta} T_{\alpha\beta} = 0$$

as claimed.

Since there is ambiguity in the choice

of  $h$ , our string theory has redundant D.O.F.

→ We can fix a gauge for the theory.

Claim: We can fix the Gauge of  $h$  in such a way that  $h$  is conformal to the flat metric in  $M^2$ .  $\exists \phi$  s.t.  
i.e.  $h_{\alpha\beta} = e^{2\phi(\sigma)} \eta_{\alpha\beta}^{(2)}$

Proof: Since  $h_{\alpha\beta} \in \text{Sym}^2(\mathbb{R})$

$h_{10} = h_{01} \Leftrightarrow \exists$  only 3 ind. param.

- Diffeomorphism invariance  $f(\sigma) \mapsto \tilde{\sigma}$  allows us to fix two of the components  $h_{10} = 0 = h_{01}$  and  $h_{00} = \pm h_{11}$

$$\Rightarrow h_{\alpha\beta} = h(x) \eta_{\alpha\beta}$$

- From Weyl invariance we can remove  $h(x)$  to obtain  $\eta_{\alpha\beta}$ !

$$\Rightarrow \text{Gauged: } \boxed{h_{\alpha\beta} = \eta_{\alpha\beta}^{(2)}}$$

- This is only true locally if the Bundle/  
Mfd are trivial & have no topological  
obstructions.

\* Geometrically: This means that the world-sheet  $\Sigma$  must have vanishing Euler-characteristic  $\chi(\Sigma)$  as

$$0 = \int d^2\sigma \sqrt{-h} R^{(2)} = \chi(\Sigma)$$

## Flat World Sheet String Action:

$$S = -\frac{T}{2} \int d^2\sigma \eta^{\alpha\beta} \partial_\alpha x^\mu \partial_\beta x^\nu$$
$$= -\frac{T}{2} \int d^2\sigma (\dot{x}^2 - \vec{x}'^2)$$

where  $\dot{x}^\mu := dx^\mu/d\sigma$ ,  $\vec{x}'^\mu = \frac{dx^\mu}{d\tau}$

\* These are the local conditions. However we can do better. Let us consider the global Criteria

$$\rightarrow h_{\alpha\beta} = e^{2\phi} \eta_{\alpha\beta}$$

$$\Leftrightarrow \delta h_{\alpha\beta} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha - 2\lambda_0 h_{\alpha\beta}$$
$$=: (P\xi)_{\alpha\beta} - 2\tilde{\lambda}_0 h_{\alpha\beta}$$

where  $P: TM \rightarrow \text{Sym}^2(TM)$  i.e.

$P$  takes a vector to a symmetric-traceless rank 2 tensor.

$$(P\xi)_{\alpha\beta} := \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha - (\nabla_\gamma \xi^\gamma) h_{\alpha\beta}$$

and

$$2\tilde{\lambda}_0 := -2\lambda_0 + \nabla_\gamma \xi^\gamma$$

Now, note that as with Killing vectors

we found that  $x^\mu \mapsto x^\mu + \xi^\mu(x) \in$

$$\Rightarrow \xi^\mu \in TM \Leftrightarrow \delta x^\mu = \mathcal{L}_\xi x^\mu$$

Here  $\delta h_{\alpha\beta} \in TM_g$ , so this variation is a tangent vector in the space

ot metrics Mg.

→ See Nakamura for a geometric view of the Teichmüller space

$$\mathcal{L}_g := \mu_g / (\text{Diff}_0(\Sigma_g) \ltimes \text{Weyl}(\Sigma_g))$$

On this tangent space we can define an inner product:

$$[\delta h^{(1)}, \delta h^{(2)}] := \int d^2\sigma \sqrt{-h} h^{\alpha\gamma} h^{\beta\delta} \delta h_{\alpha\beta}^{(1)} \delta h_{\gamma\delta}^{(2)}$$

With this, we find that  $(P^\xi)_{\alpha\beta}$  is orthogonal to the trace of  $\delta h_{\alpha\beta}$ :

$$h^{\alpha\beta} \delta h_{\alpha\beta} = 2 \nabla_r h^r - 2 \cdot 0 = \underbrace{2(\nabla_r h^r - 2\lambda_0)}_{\text{trace of } h_{\alpha\beta}.$$

trace of  $W_{\alpha\beta}$ .

thus, without loss of generality we can choose  $\lambda_D = \frac{1}{2} \nabla_\gamma h^\gamma$ , which removes the trace part + SEEMINGLY leaving the variation  $\delta h_{\alpha\beta}$  invariant since (replacing  $\nabla_\gamma h^\gamma = 2\lambda_D$ )

$$(P\xi)_{\alpha\beta} = 2(\nabla_{\alpha}\xi_{\beta}) - f_0 h_{\alpha\beta}$$

\* We conclude,  $h_{\alpha\beta} = e^{-2\phi} \hat{h}_{\alpha\beta}$  implies  $\exists$  a globally defined vector field  $\xi^\alpha$  such that

$$(P^k)_{\alpha\beta} =: L_{\alpha\beta}$$

for an arbitrary symmetric traceless  $t_{\alpha\beta}$ .

\* Note, if  $\exists$  zero modes of  $P$  i.e.

$\xi_0$  s.t.  $(P\xi_0) = 0$  then ~~is~~ solutions

• The adjoint of  $P$ ;  $P^\dagger: \text{Sym}^2(T\Sigma) \rightarrow T\Sigma$

$$(P^\dagger t)_\alpha := -2 \nabla^\beta t_{\alpha\beta}$$

The zero modes of  $P^\dagger$  are elements of  $\text{Sym}^2(T\Sigma)$  s.t.  $t_{\alpha\beta}$  cannot be written as  $(P\xi)_{\alpha\beta} \forall \xi$ .

Proof: If  $t_0$  is s.t.  $(P^\dagger t_0)_\alpha = 0$

then  $\forall \xi: (\xi, P^\dagger t_0) = 0$

$$\rightarrow (P\xi, t_0) = 0 \Rightarrow P\xi = 0$$

From  $P$  we have

$$T_g := P T_g \oplus P^\perp T_g$$

$$P T_g := \{t_{\alpha\beta} \sim (P\xi)_{\alpha\beta}\}$$

$$(P\xi) = 0 \text{ (conformal Killing eq)}$$

### Field Equations for $Sp$ :

\* Let us fix our gauge  $\hookrightarrow h_{\alpha\beta} = \eta_{\alpha\beta}$

Let  $X^m(\tau, \sigma) \mapsto X^m(\tau, \sigma) + \delta X^m(\tau, \sigma)$

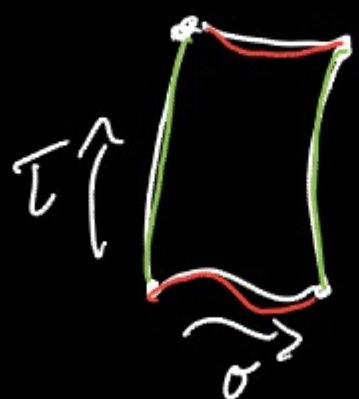
$$\delta S_p = \delta \left( \frac{T}{2} \int d\tau d\sigma (\dot{X}^2 - (X')^2) \right)$$

$$= T \int d\tau d\sigma (\dot{X}^m \delta \dot{X}_m - X'^m \delta X'_m)$$

$$= T \int d\tau d\sigma \left( \dot{X}^m \frac{d}{d\tau} (\delta X_m) - X'^m \frac{d}{d\sigma} (\delta X_m) \right)$$

$$= T \int d\tau d\sigma \left[ \frac{d}{d\tau} (\dot{X}^m \delta X_m) - \dot{X}^m \delta X_m - \frac{d}{d\sigma} (X'^m \delta X_m) + X''^m \delta X_m \right]$$

$$+ \tau \int d\sigma d\tau (\dot{x}_m^2 - \dot{x}_m^2) \delta x^m$$



$\left( \frac{\partial \tau}{\partial \sigma} \right)$  It's clear we must Not have a tangent vector at the boundary of  $\partial \tau$

• For the classical particle we have



and  $\delta x_1 = \delta x_2 = 0$

$\forall t$ , meaning the worldsheet is fixed for that time.

Wouldn't make sense to allow for varying end points when maximizing  $S_p$ .

$$\therefore \left[ \int d\sigma (\dot{x}_m \delta x^m) \right]_{\partial \tau} = 0$$

$$\Rightarrow \delta S = 0$$

$$\neq 0$$

$$\left( -\partial_\tau^2 + \partial_\sigma^2 \right) x^m - \tau \int d\sigma \left[ x_n' \delta x^m \Big|_{\sigma=0} - x_n' \delta x^m \Big|_{\sigma=l} \right]$$

\* This Boundary

Boundary term (B)

term determines the type of string we have.

## (1) Closed String:

For a closed string we take  $\sigma \sim \sigma + l$

$$x^m(\tau, \sigma + l) = x^m(\tau, \sigma)$$

$$\delta X^\mu(\bar{\tau}, \sigma + l) = \delta X^\mu(\bar{\tau}, \sigma) \Rightarrow \textcircled{B} = 0,$$

This leaves

$$\boxed{(\partial_{\bar{\tau}}^2 + \partial_{\sigma}^2) X^\mu(\bar{\tau}, \sigma) = 0}$$

$$\square X^\mu(\bar{\tau}, \sigma) = 0$$

$$\boxed{\text{B.C.'s: } X^\mu(\bar{\tau}, \sigma + l) = X^\mu(\bar{\tau}, \sigma)}$$

Open strings: There are two possibilities

$$\partial_{\sigma} X^\mu|_{\sigma=0,l} = 0 \quad (\text{Neumann B.C.'s})$$

$$\delta X^\mu|_{\sigma=0,l} = 0 \quad (\text{Dirichlet B.C.'s})$$



$\sigma=0$

Neumann: The string can oscillate and its endpoints can move along  $\partial\sigma$  as long as the vectors  $\boxed{\partial_{\sigma} X^\mu = 0}$  at  $\partial\sigma$



$\sigma=0$

$\sigma=l$

Dirichlet: String can oscillate but its end points are fixed

$$\delta X^\mu|_{\sigma=0,l} = 0$$

$$x'^{\mu} - x^{\mu} = \delta x^{\mu}$$

This is why  $\delta X^\mu|_{\partial\sigma} = 0$  for Dirch since  $\uparrow = 0$  at  $\partial\sigma$ .

Poincare:

- Neumann: These B.C.'s  $\partial_\sigma X^\mu / \partial \sigma = 0$  are preserved under Poincaré

$$\begin{aligned}\partial_\sigma (X^\mu) |_{\sigma=0,e} &= \partial_\sigma (\lambda^\mu_\nu X^\nu + a^\mu) |_{\sigma=0,e} \\ &= \lambda^\mu_\nu \partial_\sigma X^\nu |_{\sigma=0,e} = 0 \quad \checkmark\end{aligned}$$

• Dirichlet:

Let us be slightly more concrete.

Since  $\delta X^\mu(\tau, 0) = \delta X^\mu(\tau, e) = 0$

it's clear that  $X^\mu / \partial \sigma$  is

constant. So we define the following

$$X^\mu(\tau, 0) =: X^\mu_0$$

$$X^\mu(\tau, e) =: X^\mu_e$$

Although  $X^\mu_0 = X^\mu_0(\tau)$  it's clear that this does not vary in time either so this really is a constant.

Claim: D. B.C.'s do not preserve the IMP Poincaré symmetry:

Proof:

$$\begin{aligned}X^\mu |_{\sigma=0,e} &= (\lambda^\mu_\nu X^\nu + b^\mu) |_{\sigma=0,e} \\ &= \lambda^\mu_\nu (X^\nu) |_{\sigma=0,e} + b^\mu \\ &\neq X^\mu_{0,\mu}\end{aligned}$$

\* Dirichlet boundary conditions give rise to  $D_p$ -branes.

Suppose we have Neumann B.C.'s

coord's.

$\partial_\sigma X^\mu|_{\sigma=0,L} = 0$ ,  $\mu = 1, \dots, p+1$   
and Dirichlet B.C.'s on  $D-(p+1)$   
coords  $X^\nu|_{\sigma=0,L} \neq \nu = 0, 1, \dots, D$

$\Rightarrow$  The string ends on a  $P_p$ -brane.

Solving the field Eqs:  $(\partial_\sigma^2 - \partial_\tau^2)X^\mu = 0$

• Because we have gauge fixed  $\eta_{\alpha\beta} = \eta_{\alpha\beta}$   
we must impose the tracelessness  
of the E.M. tensor

$$\eta^{\alpha\beta} T_{\alpha\beta} = 0 \Leftrightarrow \partial_\alpha X^\mu \partial_\beta X_\mu = \frac{1}{2} \eta_{\alpha\beta} \eta^{\delta\delta} \partial_\delta X^\mu \partial_\delta X_\mu$$

$$\text{or } T_{00} = T_{11} = \frac{1}{2} (\dot{X}^2 + X'^2) = 0$$

$$T_{01} = T_{10} = \dot{X}_\mu X'^\mu = 0$$

light-cone coordinates:  $\sigma^\pm := \tau \pm \sigma$

$$\sigma = \frac{1}{2}(\sigma^+ - \sigma^-), \quad \tau = \frac{1}{2}(\sigma^+ + \sigma^-)$$

$$\partial_\tau = \partial/\partial\sigma^+ = \frac{\partial\tau}{\partial\sigma^+} \frac{\partial}{\partial\tau} + \frac{\partial\sigma}{\partial\sigma^+} \frac{\partial}{\partial\sigma}$$

$$\Rightarrow \frac{1}{2}(\partial_\tau + \partial_\sigma) = \partial_+$$

$$\partial_- = \frac{1}{2}(\partial_\tau - \partial_\sigma)$$

$$\eta_{\pm\pm} = -\frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\eta_{+-} = -1/2, \quad \eta_{++} = 0$$

$$\eta_{-+} = -1/2, \quad \eta_{--} = 0$$

$$\eta^{\pm\pm} = -2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$\therefore$  The E.O.M becomes

$$(-\partial_\sigma^2 + \partial_\tau^2)X^\mu = 0$$

$$(\partial_+ \partial_-)X^\mu = 0$$

$$\partial_- \partial_+ X^m = 0 \quad (a)$$

$$\Rightarrow T_{++} = \partial_+ X^m \partial_+ X_m = 0 \quad (b)$$

$$T_{--} = \partial_- X^m \partial_- X_m = 0 \quad (c)$$

• a, b, c are the equations we must solve for.

(Not in many textbooks)

### Proof of mode expansion for Closed Strings:

We start in  $1M^2 \cong \Sigma$  and  $(\partial_\sigma^2 - \partial_\tau^2)X^m = 0$

(i) Assume  $X^m(\tau, \sigma) = g(\tau) f^m(\sigma)$

$$\Leftrightarrow \left( \frac{d^2 f^m(\sigma)}{d\sigma^2} \right) - \frac{f^m(\sigma)}{g(\tau)} \frac{d^2 g(\tau)}{d\tau^2} = 0$$

$$\Leftrightarrow \exists c \in \mathbb{R} \text{ s.t. } \frac{d^2 f^m(\sigma)}{d\sigma^2} = c f^m(\sigma) \text{ and } \frac{d^2 g(\tau)}{d\tau^2} = c g(\tau)$$

We assume that  $c = -\lambda^2$

$$\frac{d^2 f^m}{d\sigma^2} = -\lambda^2 f^m(\sigma) \quad \text{and} \quad \dots$$

$$f^m(\sigma) = A^m \cos(\sigma) + B^m \sin(\sigma)$$

$$g(\tau) = A \cos(\tau) + B \sin(\tau)$$

We must now impose Closed String B.C.'s

$$\Rightarrow f^m(0) = f^m(l) = A^m \cos(l) + B^m \sin(l) = A^m$$

$$\Rightarrow \sin(l\lambda) = 0 \Leftrightarrow l\lambda = n\pi, \quad n \in \mathbb{Z}$$

$$\lambda = \frac{n\pi}{l}$$

However we must also have

$$\cos(l\lambda) = 1 \Rightarrow l\lambda = 2\pi n$$

$$\lambda = 2\pi n / l$$

Likewise  $g(\tau) = A \cos(2\pi n \tau / \ell) + B \sin(2\pi n \tau / \ell)$

→ Go to  $e^{\pm i}$ :

$$f_n^\mu(\sigma) = \tilde{A}^\mu e^{-i \frac{2\pi n}{\ell} \sigma} + \tilde{B}^\mu e^{i \frac{2\pi n}{\ell} \sigma}$$

$$g(\tau) = \tilde{A} e^{-i \frac{2\pi n}{\ell} \tau} + \tilde{B} e^{i \frac{2\pi n}{\ell} \tau}$$

$$\Leftrightarrow (f_n^\mu g_n) = x_{(n)}^\mu = \tilde{A}^\mu \tilde{A} e^{-i\lambda_n(\sigma+\tau)} + \tilde{B}^\mu \tilde{B} e^{i\lambda_n(\sigma+\tau)} + \tilde{B}^\mu \tilde{A} e^{-i\lambda_n(\tau-\sigma)} + \tilde{A}^\mu \tilde{B} e^{i\lambda_n(\tau-\sigma)}$$

check

$$x^\mu(0,0) = x^\mu(\tau,\ell)$$

$$\tilde{A}^\mu e^{i\lambda_n \tau} + \tilde{B}^\mu e^{i\lambda_n \tau} + C^\mu e^{-i\lambda_n \tau} + D^\mu e^{i\lambda_n \tau}$$

$$= e^{-i\lambda_n \tau} (A^\mu + C^\mu) + e^{i\lambda_n \tau} (B^\mu + D^\mu)$$

→ Conclusion:

$$f_n^\mu(\sigma) g_n(\tau) = A^\mu e^{-i\lambda_n(\tau+\sigma)} + B^\mu e^{-i\lambda_n(\tau-\sigma)}$$

$n \neq 0$

For  $n=0 \Rightarrow \lambda=0$

$$\frac{\partial^2 x^\mu}{\partial \sigma^2} = 0 \Rightarrow a^\mu \sigma + b^\mu = f_{(0)}^\mu(\sigma)$$

$$f_{(0)}^\mu(\ell) = a^\mu \ell + b^\mu = b^\mu \Rightarrow a^\mu = 0$$

$$\therefore f_{(0)}^\mu(\sigma) = 0$$

$$g(\tau) = a\tau + b \quad (\text{no restrictions})$$

$$\begin{aligned} x^\mu(\sigma, \tau) &= b^\mu (a\tau + b) + \sum_{n \neq 0} \frac{1}{n} A_n^\mu e^{-i\lambda_n(\tau+\sigma)} + \sum_{n \neq 0} \frac{1}{n} B_n^\mu e^{-i\lambda_n(\tau-\sigma)} \\ &= b^\mu b + ab^\mu \tau + \sum_{n \neq 0} \left( \right) \end{aligned}$$

$$\therefore b^\mu b =: x^\mu$$

$$ab^\mu =: \ell_s^2 p^\mu$$

$$X^\mu(\sigma, \tau) = x^\mu + \tau l_s^2 p^\mu + \sum_{n \neq 0} A_n^\mu e^{-in(\tau+\sigma)} + B_n^\mu e^{-in(\tau-\sigma)}$$

Use last conditions

$$\dot{X}^\mu = l_s^2 p^\mu + \sum_{n \neq 0} A_n^\mu (-in) e^{-in(\tau+\sigma)} + (-in) B_n^\mu e^{-in(\tau-\sigma)}$$

$$X'^\mu = \sum_{n \neq 0} A_n^\mu (-in) e^{-in(\tau+\sigma)} + B_n^\mu (in) e^{-in(\tau-\sigma)}$$

$$X^\mu = X_R^\mu + X_L^\mu = \underbrace{x^\mu + \tau l_s^2 p^\mu}_{\text{Center of mass}} + \underbrace{\frac{i}{2} l_s^2 \sum_{n \neq 0} \frac{1}{n} (\alpha_n^\mu e^{-in(\tau+\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau-\sigma)})}_{\text{Oscillation of string.}} e^{-2in\tau}$$

$$\tau = \frac{1}{2\pi\alpha'}, \quad \alpha' = \frac{1}{2} l_s^2$$

$$X_R^\mu = \frac{1}{2} X^\mu + \frac{1}{2} l_s^2 (\tau - \sigma) p^\mu + \frac{i}{2} l_s^2 \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in(\tau-\sigma)}$$

$$X_L^\mu = \frac{1}{2} X^\mu + \frac{1}{2} l_s^2 (\tau + \sigma) p^\mu + \frac{i}{2} l_s^2 \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)}$$

- Note the first term looks like the trajectory of a point particle

$$\frac{d^2}{d\tau^2} (X^\mu) = 0 \Rightarrow X^\mu = x^\mu + p^\mu \tau$$

$\Rightarrow X^\mu$  must be real

$$\Rightarrow x^\mu, p^\mu \in \mathbb{R}$$

$$\text{Note: } \left( \sum_{n \neq 0} \frac{i}{n} (\alpha_n^\mu)^* e^{in(\tau-\sigma)} \right)$$

let  $n \mapsto -n$ , then

$$\sum_{n \neq 0} \frac{i}{n} (\alpha_{-n}^\mu)^* e^{-in(\tau-\sigma)}$$

$$\Leftrightarrow (\alpha_{-n}^\mu)^* = \alpha_n^\mu$$

$$\tilde{\alpha}_{-n}^\mu = (\tilde{\alpha}_n^\mu)^*$$

• Recall  $\Pi(\tau, \sigma) = \partial \dot{X}^\mu = T \dot{X}^\mu = \frac{1}{\pi \alpha'} \dot{X}^\mu$   
 $= \frac{p^\mu}{\pi} + \frac{1}{\pi \alpha'} \sum_{n \neq 0} (\alpha_n^\mu e^{-in\tau(\tau+\sigma)} + \tilde{\alpha}_n^\mu e^{-in\tau(\tau-\sigma)})$

\* We find this momenta satisfy the following

Poisson bracket:

$$\{ \Pi^\mu(\tau, \sigma), \Pi^\nu(\tau, \sigma') \}_{P.B.} = 0$$

$$\{ X^\mu(\tau, \sigma), X^\nu(\tau, \sigma') \}_{P.B.} = 0$$

$$\{ \Pi^\mu(\tau, \sigma), X^\nu(\tau, \sigma') \}_{P.B.} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

$$\Rightarrow \{ \alpha_m^\mu, \alpha_n^\nu \}_{P.B.} = \{ \tilde{\alpha}_m^\mu, \tilde{\alpha}_n^\nu \}_{P.B.} = i m \eta^{\mu\nu} \delta_{m, -n}$$

$$\{ \alpha_m^\mu, \tilde{\alpha}_n^\nu \}_{P.B.} = 0$$

$$\{ p^\mu, x^\nu \}_{P.B.} = \eta^{\mu\nu}$$

Proof:  $p^\mu(\tau, \sigma) = \frac{p^\mu}{\pi} + \frac{1}{\pi \alpha'} \sum_{n \neq 0} (\alpha_n^\mu e^{-in\tau(\tau+\sigma)} + \tilde{\alpha}_n^\mu e^{-in\tau(\tau-\sigma)})$

$$p^\mu(\tau, \sigma) p^\nu(\tau, \sigma') = \left( \frac{p^\mu p^\nu}{\pi^2} + p^\mu \frac{1}{\pi \alpha'} \sum_{n \neq 0} (\alpha_n^\nu e^{-in\tau(\tau+\sigma')} + \tilde{\alpha}_n^\nu e^{-in\tau(\tau-\sigma')}) \right. \\ \left. + \left( \frac{1}{\pi \alpha'} \right)^2 \sum_{n \neq 0} (\alpha_n^\mu e^{-in\tau(\tau+\sigma)} + \tilde{\alpha}_n^\mu e^{-in\tau(\tau-\sigma)}) \sum_{m \neq 0} (\alpha_m^\nu e^{-im\tau(\tau+\sigma')} + \tilde{\alpha}_m^\nu e^{-im\tau(\tau-\sigma')}) \right)$$

\* For simplicity:  $\sigma' = 0$ : Project the eigen basis of  $\frac{\partial^2}{\partial \sigma^2}$  (i.e. do

Fourier transform:

$$p^\mu(\tau, \sigma) p^\nu(\tau, 0) = \frac{p^\mu p^\nu}{\pi^2} + \frac{p^\mu}{\pi \alpha'} \sum_{n \neq 0} (\alpha_n^\nu e^{-in\tau} + \tilde{\alpha}_n^\nu e^{-in\tau}) + \frac{p^\nu}{\pi \alpha'} \sum_{m \neq 0} (\alpha_m^\mu e^{-im\tau} + \tilde{\alpha}_m^\mu e^{-im\tau}) \\ + \left( \frac{1}{\pi \alpha'} \right)^2 \sum_{n \neq 0} \sum_{m \neq 0} (\alpha_n^\mu e^{-in\tau} + \tilde{\alpha}_n^\mu e^{-in\tau}) (\alpha_m^\nu e^{-im\tau} + \tilde{\alpha}_m^\nu e^{-im\tau})$$

$$+ m \rightarrow n \quad \div 0$$

$$\bullet \sum_n \sum_m \left[ \alpha_n^\mu \alpha_m^\nu e^{-in(2\tau-\sigma)} + \tilde{\alpha}_n^\mu \alpha_m^\nu e^{-in(2\tau+\sigma)} + \alpha_n^\mu \tilde{\alpha}_m^\nu e^{-in(2\tau-\sigma)} + \tilde{\alpha}_n^\mu \tilde{\alpha}_m^\nu e^{-in(2\tau+\sigma)} \right]$$

$$(ii) \{ X^\mu(\tau, \sigma), X^\nu(\tau, 0) \}_{P.B.} \stackrel{!}{=} 0$$

term:

$$-\frac{1}{2} \alpha'^2 \sum_n \sum_m (\alpha_n^\mu e^{-in\tau} + \tilde{\alpha}_n^\mu e^{-in\tau}) (\alpha_m^\nu + \tilde{\alpha}_m^\nu) e^{-4i(n+m)\tau}$$

$$= -\frac{1}{2} \alpha'^2 \sum_n \sum_m (\alpha_n^\mu \alpha_m^\nu e^{-in\tau} + \tilde{\alpha}_n^\mu \alpha_m^\nu e^{-in\tau} + \alpha_n^\mu \tilde{\alpha}_m^\nu e^{-in\tau} + \tilde{\alpha}_n^\mu \tilde{\alpha}_m^\nu e^{-in\tau}) e^{-4i(n+m)\tau}$$

$$\{P^\mu(\tau, \sigma), P^\nu(\tau, 0)\}_{P,B} = 0 \Leftrightarrow$$

$$\left\{ p^\mu + \frac{1}{\pi \alpha'} \sum_n \left( \alpha_n^\mu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-2in(\tau+\sigma)} \right), \frac{p^\nu}{\pi} + \frac{1}{\pi \alpha'} \sum_n \left( \alpha_n^\nu e^{-2in(\tau-\sigma)} + \tilde{\alpha}_n^\nu e^{-2in(\tau+\sigma)} \right) \right\}$$

$$\frac{1}{\pi} \{p^\mu, p^\nu\} + \frac{1}{\pi \alpha'} \sum_n \left\{ p^\mu, \alpha_n^\nu \right\} e^{-2in(\tau)} + \{p^\mu, \tilde{\alpha}_n^\nu\} e^{-2in(\tau)}$$

$$+ \frac{1}{\pi \alpha'} \sum_n \left\{ \alpha_n^\mu, p^\nu \right\} e^{-2in(\tau-\sigma)} + \{ \tilde{\alpha}_n^\mu, p^\nu \} e^{-2in(\tau+\sigma)}$$

$$+ \frac{1}{(\pi \alpha')^2} \sum_{n,m} \left( \{ \alpha_n^\mu, \alpha_m^\nu \} e^{-2in(2\tau-\sigma)} + \{ \alpha_n^\mu, \tilde{\alpha}_m^\nu \} e^{-2in(2\tau-\sigma)} + \{ \tilde{\alpha}_n^\mu, \alpha_m^\nu \} e^{-2in(2\tau+\sigma)} \right.$$

$$\left. + \{ \tilde{\alpha}_n^\mu, \tilde{\alpha}_m^\nu \} e^{-2in(2\tau+\sigma)} \right)$$

→ Set each term to zero:

$$\{p^\mu, p^\nu\} = 0$$

$$\{ \alpha_n^\mu, p^\nu \} e^{-2in\tau} + \{ \tilde{\alpha}_n^\mu, p^\nu \} e^{2in\tau} = 0$$

$$\{ p^\mu, \alpha_m^\nu \} e^{-2im\tau} + \{ p^\mu, \tilde{\alpha}_m^\nu \} e^{2im\tau} = 0$$

$$\{ \alpha_n^\mu, \alpha_m^\nu \} e^{-2i(n+m)\tau} + \dots = 0$$

→ Simply equate coefficients of terms  $e^{in\sigma}$  (and  $\sigma$ ) etc..

•  $f_n(\sigma) := e^{in\sigma}/\sqrt{\pi}$ ,  $n \in \mathbb{Z}$  form a complete basis for  $f^\mu$  defined for  $\sigma \in [0, \pi]$ . Any  $f^\mu(\sigma, \sigma') \sim \sum_n f_n(\sigma) f_n(\sigma')$

• Similarly for  $\{x^\mu(\tau, \sigma), x^\nu(\tau, 0)\} = 0$  &  $\{p^\mu(\tau, \sigma), x^\nu(\tau, 0)\} = \eta^{\mu\nu}$ .

$$\therefore \{p^\mu, p^\nu\} = \{x^\mu, x^\nu\} = 0, \quad \{p^\mu, x^\nu\} = \eta^{\mu\nu}$$

$$\{ \alpha_n^\mu, \alpha_m^\nu \} = \{ \tilde{\alpha}_n^\mu, \tilde{\alpha}_m^\nu \} = i\alpha' \eta^{\mu\nu} \delta_{n+m, 0}, \quad \{ \alpha_n^\mu, \tilde{\alpha}_m^\nu \} = 0$$

\* The classical Poisson bracket is defined by a symplectic structure of the embedded  $M^{2d}$

$$\{F, G\} := \int d\sigma \left( \frac{\partial F(\tau, \sigma)}{\partial x^\mu(\tau, \sigma)} \frac{\partial G(\tau, \sigma)}{\partial \pi_\mu(\tau, \sigma)} - \frac{\partial G(\tau, \sigma)}{\partial x^\mu(\tau, \sigma)} \frac{\partial F(\tau, \sigma)}{\partial \pi_\mu(\tau, \sigma)} \right)$$

The Hamiltonian is given by  $H = \int_0^l d\sigma \underbrace{\left( \dot{x}^\mu(\tau, \sigma) \pi_\mu(\tau, \sigma) - \mathcal{L} \right)}_{\text{Hamiltonian Density} \rightarrow \mathcal{H}}$

Open strings:

→ These give a much different solution space

$$X^\mu(\tau, \sigma) = f(\sigma) g(\tau) \quad \Leftrightarrow \quad \frac{\partial^2 f(\sigma)}{\partial \sigma^2} \frac{1}{f} = \lambda$$

$$f(\sigma) = e^{\pm i\pi\sigma}, \quad g(\tau) = e$$

$$\partial_\sigma f(\sigma) = \pm i\pi e^{\pm i\pi\sigma} \Big|_{0, \ell} = 0$$

$$|_0 : \lambda = 0$$

$$f(\sigma) = a^\mu \sigma + b^\mu$$

$$|_{\sigma=\ell} : \partial_\sigma f^\mu(\sigma) = A^\mu e^{i\pi\sigma} + B^\mu (-i\pi) e^{-i\pi\sigma}$$

$$\Leftrightarrow A^\mu e^{i\pi\ell} + B^\mu e^{-i\pi\ell} = 0$$

$$B = A^\mu e^{2i\pi\ell}$$

$$\text{Assuming } \frac{\partial^2 f}{\partial \sigma^2} = -\lambda^2$$

$$f^\mu(\sigma) = A^\mu \cos(\lambda\sigma) + B^\mu \sin(\lambda\sigma)$$

$$\partial_\sigma f^\mu(\sigma) = -\lambda A^\mu \sin(\lambda\sigma) + \lambda B^\mu \cos(\lambda\sigma)$$

$$|_{\sigma=0} : \lambda B^\mu = 0$$

$$\bullet \lambda = 0 \Rightarrow \underline{f^\mu(\sigma) = a^\mu \sigma + b^\mu}$$

$$\bullet B^\mu = 0 \quad \underline{f^\mu(\sigma) = A^\mu \cos(\lambda\sigma)}$$

$$\text{For } \partial_\sigma X^\mu(\tau, \sigma)|_{\sigma=\ell} = 0$$

$$\Leftrightarrow \underline{f_{\cos}^\mu(\sigma) = A^\mu}$$

$$\Leftrightarrow \sin(\lambda\sigma) = 0 \Leftrightarrow \lambda \ell = \pi n, \quad n \in (\mathbb{Z} \setminus \{0\})$$

$$\underline{f_{\cos}^\mu(\sigma) = A^\mu \cos\left(\frac{\pi n}{\ell} \sigma\right)}$$

$$\text{w/ No constraints on } g(\tau) = A\tau + b$$

$$g_{\cos}(\tau) = (A e^{-i\pi\tau} + B e^{i\pi\tau})$$

$$X^\mu(\tau, \sigma) = a^\mu (A\tau + b) \quad (\text{See 4.2 for Dirichlet}) \dots$$

## Symmetries in Canonical Quantization:

• For a global symmetry of the system on the worldsheet

$$\exists j^a : \quad \partial_\tau j^a = 0 \quad \Leftrightarrow \quad \frac{d}{dt} Q = \frac{d}{dt} \left( \int d\sigma j^0 \right) = 0$$

$$\frac{d}{dt} \left( \int d\sigma j^0 \right) = \int d\sigma \frac{d}{dt} j^0$$

$$\frac{d\alpha}{dt} = - \int_0^1 d\sigma \frac{\partial}{\partial \tau} j^\sigma = - j^\sigma|_{\sigma=0}^{\sigma=1} = 0$$

For  $U: \phi \mapsto \phi + \delta\phi$ ,  $\delta\phi \in U(\phi)$

Take  $\epsilon$  to be local, then

$$\delta S = \int d\tau d\sigma (\partial_\alpha \epsilon) j^\alpha = - \int d\tau d\sigma \epsilon \partial_\alpha j^\alpha$$

(i) Poincaré Charge/Current for Bosonic string:

(a) Translations  $x^\mu \mapsto x^\mu + a^\mu(\sigma)$

$$S_p = -\frac{T}{2} \int d\tau d\sigma \eta^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu$$

$$\hookrightarrow \delta S_p = -T \int d\tau d\sigma \partial_\alpha (a^\mu(\sigma)) \partial^\alpha X_\mu$$

$$\Rightarrow \boxed{j^\alpha_\mu = -T \partial^\alpha X_\mu}$$

This implies the current can be written as

$$j^\alpha_\mu = -T \partial^\alpha \eta_{\mu\nu} X^\nu \Leftrightarrow \underline{j^{\alpha\mu} = -T \partial^\alpha X^\mu}$$

Interpret this as  $\exists$   $\mu$  currents  $j^\alpha$  for each  $\mu$   $X^\mu$  which embed the string.

To see that this current is conserved:

$$\partial_\alpha j^{\alpha\mu} = -T \partial_\alpha \partial^\alpha X^\mu = -T (-\partial_\tau^2 + \partial_\sigma^2) = 0$$

The conserved charge  $Q = \int d\sigma j^0$  is naturally going to be the string momentum

$$P^\mu = \int d\sigma (-T \partial^0 X^\mu) = T \int d\sigma \partial_\tau X^\mu = T \int d\sigma \dot{X}^\mu = \int d\sigma T P^\mu$$

$P^\mu$ : Canonical momentum, conjugate to  $X^\mu \Rightarrow P^\mu = \Pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu}$

Thus, we see that  $P^\mu$  in the  $X^\mu(\tau, \sigma)$  was chosen to

show that the action  $S_\sigma$  has translational symmetry.

(b) Lorentz transformations:  $x^\mu \mapsto (\Lambda^{-1})^\mu_\nu x^\nu$

$$\Rightarrow (\Lambda^{-1})^\mu_\nu = (\Lambda^{-1})^\mu_\nu(x)$$

$$\delta S_p = -T \int d\sigma d\tau \partial_\alpha ((\Lambda^{-1})^\mu_\nu X^\nu) \partial^\alpha X_\mu$$

$$= -T \int d\sigma d\tau \left[ \partial_\alpha ((\Lambda^{-1})^\mu_\nu) X^\nu \partial^\alpha X_\mu + (\Lambda^{-1})^\mu_\nu (\partial_\alpha X^\nu) \partial^\alpha X_\mu \right]$$

The second term is:

$$(\Lambda^{-1})^\mu_\nu (\partial_\alpha X^\nu) \partial^\alpha X_\mu = 0$$

$$\begin{array}{c} \xrightarrow{\text{Anti}} \quad \xrightarrow{\text{Sym}} \\ \Rightarrow \mathcal{L}_P = -T \int d\sigma d\tau \partial_\alpha (\Delta^{-1})^\mu_\nu X^\nu \partial^\alpha X_\mu = \int d\sigma d\tau (\partial^\alpha \Delta_{\mu\nu}) X^\mu \partial_\alpha X^\nu \\ \therefore j_\alpha^{\mu\nu} = -\frac{T}{2} (X^\mu \partial_\alpha X^\nu - X^\nu \partial_\alpha X^\mu) \end{array}$$

Q: How do we know to "make"  $j_\alpha^{\mu\nu}$  antisymmetric? Note that  $\Delta_{\mu\nu}$  is antisym and hence so too should the current.

- $j_\alpha^{\mu\nu}$  is only conserved on-shell.

### Hamiltonian & E.M. Tensor:

- Hamiltonian density  $\mathcal{H} = \dot{X}_\mu p^\mu - \mathcal{L}$

$$\text{Hamiltonian: } H := \int_0^l d\sigma (\mathcal{H})$$

$$H = \int_0^l d\sigma (\dot{X}^\mu p_\mu - \mathcal{L})$$

For the Bosonic string  $p^\mu = T \dot{X}^\mu$  and  $\Delta \cong M^2: \mathcal{L} = \frac{1}{2} \dot{X}^2$ .

$$S = \int d\tau d\sigma \left( -\frac{T}{2} \right) (-2 \dot{X}^\mu \partial_\tau X_\mu + \partial_\sigma X^\mu \partial_\sigma X_\mu)$$

$$H = \int d\sigma \left( T \dot{X}^2 - \frac{T}{2} \dot{X}^2 + \frac{T}{2} X'^2 \right) = \frac{T}{2} \int d\sigma (\dot{X}^2 + X'^2)$$

$$\therefore H = \frac{T}{2} \int d\sigma (\dot{X}^2 + X'^2) = T \int_0^l d\sigma ((\partial_\tau X)^2 + (\partial_\sigma X)^2)$$

- To understand how to interpret  $H$  we must expand  $X_\mu^h$  its oscillation modes.

### Closed string Hamiltonian:

$$\dot{X}^\mu = l_s^2 p^\mu + l_s \sum_{n \neq 0} (\alpha_n^\mu e^{2in\sigma} + \tilde{\alpha}_n^\mu e^{-2in\sigma}) e^{-2in\tau}$$

$$\therefore \dot{X}^\mu \dot{X}_\mu = (l_s^2 p^2 + 2l_s p^\mu \sum_n (\alpha_n^\mu + \tilde{\alpha}_n^\mu) e^{2in\sigma} + \sum_{m,n} (\alpha_m^\mu \alpha_n^\mu + \tilde{\alpha}_m^\mu \tilde{\alpha}_n^\mu) e^{2i(m+n)\sigma}) e^{-2i(m+n)\tau} (\dots)$$

$$X'^\mu = -l_s \sum_n (\alpha_n^\mu e^{2in\sigma} - \tilde{\alpha}_n^\mu e^{-2in\sigma}) e^{-2in\tau}$$

$$X'^2 \cong l_s^2 \sum_n \sum_m (\alpha_n^\mu \alpha_m^\mu e^{2i(n+m)\sigma} + \dots)$$

$\therefore$  we get terms such as

$$\int_0^l d\sigma \alpha_n^\mu (\alpha_m)_\mu e^{2i(n+m)\sigma} + \dots$$

However we know that  $\int d\sigma e^{2i(n+m)\sigma} = l \delta(n+m)$

This is due to the fact that the length of the string

to  $\int_{-\infty}^{\infty} d\sigma$ .

\*  $H = \frac{1}{2} \sum_{-\infty}^{\infty} (\alpha_n \cdot \alpha_n + \tilde{\alpha}_n \cdot \tilde{\alpha}_n)$ ,  $\alpha_0^\mu := \tilde{\alpha}_0^\mu = \frac{1}{2} l_s p^\mu$

Open strings:  $H = \frac{1}{2} \sum_{-\infty}^{\infty} (\alpha_n \cdot \alpha_n)$ ,  $\alpha_0^\mu = \tilde{\alpha}_0^\mu = l_s p^\mu$

Trivially  $H$  is conserved  $\frac{d}{d\tau}(H) = 0$

\* Stress-Eu. tensor (Closed String):

In light-cone coordinates:  $T_{--} = (\partial_- X_\mu)^2$ ,  $T_{++} = (\partial_+ X_\mu)^2$

$T_{+-} = T_{-+} = 0$

$\tau = (\sigma^+ + \sigma^-) \frac{1}{2}$ ,  $\sigma = (\sigma^+ - \sigma^-) \frac{1}{2}$

Proof:  $T_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} \eta_{\alpha\beta} \eta^{\mu\nu} \partial_\mu X^\nu \partial_\nu X_\mu = 0$

$T_{++} = \partial_+ X^\mu \partial_+ X_\mu - \frac{1}{2} \eta^{++} \partial_\mu X^\mu \partial_\nu X_\nu = 0$

$T_{+-} = \partial_+ X^\mu \partial_- X_\mu - \partial_+ X^\mu \partial_- X_\mu = 0$

$X^\mu(\sigma, \tau) = X_R^\mu(\sigma^-) + X_L^\mu(\sigma^+)$

$\Rightarrow T_{++} = (\partial_+ X_L^\mu \partial_+ X_{L\mu}) = 0$

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$\Rightarrow T_{++} = \left( \frac{1}{2} l_s^2 p^\mu p_\mu + l_s^2 \sum_{n \neq 0} \tilde{\alpha}_n^\mu \tilde{\alpha}_{-n\mu} e^{-2in(\tau-\sigma)} \right)$   
 $= l_s^2 \alpha_0^\mu \alpha_{0\mu} + l_s^2 \sum_{n \neq 0} \tilde{\alpha}_n^\mu \tilde{\alpha}_{-n\mu} e^{-2in(\tau-\sigma)}$   
 $= l_s^2 \left( \sum_{-\infty}^{\infty} \sum_{-\infty}^{\infty} \tilde{\alpha}_m^\mu \tilde{\alpha}_{n\mu} e^{-2i(m+n)(\tau-\sigma)} \right)$

Let  $m+n = p$

$T_{++} = l_s^2 \sum_{-\infty}^{\infty} \sum_{-\infty}^{\infty} \tilde{\alpha}_{p-n}^\mu (\tilde{\alpha}_n)_\mu e^{-2ip(\tau-\sigma)}$

we define

$= 2 l_s^2 \sum_{-\infty}^{\infty} L_m e^{-2im(\tau-\sigma)}$

$L_m := \left( \frac{1}{2} \right) \sum_{-\infty}^{\infty} \tilde{\alpha}_{m-n} \cdot \tilde{\alpha}_n$  (Classical)

$T_{--} = (\partial_- X_\mu)^2$

$= 2 l_s^2 \sum_{-\infty}^{\infty} L_m e^{-2im(\tau-\sigma)}$

$L = \frac{1}{2} \sum_{-\infty}^{\infty} \alpha_{m-n} \cdot \alpha_n$  (Classical)

we will show that  $L_n$  are the conserved charges associated to conformal Killing fields.

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[The following text is a dense, illegible block of characters and symbols, likely representing a corrupted or redacted document. It contains no discernible words or structure.]